

Inferring Planetary Companions from Gaia RUWE: A PyMC-Based Bayesian Astrometric Pipeline

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Abstract

The Gaia space mission provides unprecedented astrometric precision for over one billion stars. The Renormalised Unit Weight Error (RUWE), a statistic published in Gaia Data Release 3, measures the excess astrometric scatter around a single-star fit and serves as a sensitive indicator of unresolved companions. In this independent study, I developed a pipeline for inferring the properties of planetary and substellar companions from RUWE using PyMC-based Bayesian inference. The pipeline comprises four stages: data acquisition from the Gaia DR3 archive and scanning-law computation (Part 1), PyTensor-compatible forward modeling of Gaia along-scan positions for both single-star and planet-companion scenarios (Part 2), model-predicted RUWE calculation (Part 3), and full posterior sampling of orbital and astrometric parameters using PyMC (Part 4). I compare this implementation against the reference Stan-based pipeline (gaia-astro-fit) with respect to accuracy, performance, and extensibility. The new pipeline successfully samples posteriors with no divergences, acceptable R-hat values, and effective sample sizes exceeding 400 per parameter, and is architected to take full advantage of Gaia Data Release 4 epoch astrometry when it becomes available.

1. Background on Gaia and Astrometric Planet Detection

1.1 The Gaia Mission

Gaia is a space astrometry mission operated by the European Space Agency (ESA), launched in December 2013 and placed at the Sun-Earth L2 Lagrange point. Its primary scientific objective is to construct the most precise three-dimensional map of the Milky Way by measuring the positions, parallaxes, and proper motions of approximately 1.5 billion stars. The satellite carries two optical telescopes, two fields of view separated by 106.5° , and a focal plane with 106 CCDs operating in

time-delayed integration mode. Gaia observes each source multiple times over its mission lifetime with each transit producing roughly 9 individual along-scan (AL) measurements.

As of Gaia Data Release 3 (DR3, June 2022), the archive contains five-parameter astrometric solutions position, parallax, and proper motion for 1.47 billion sources, with a single-measurement along-scan precision ranging from 0.03 mas for $G < 13$ to tens of mas for $G > 19$. The formal uncertainties on parallax and proper motion for bright stars ($G < 14$) are at the few-microarcsecond level, making Gaia the most precise astrometric instrument ever operated.

1.2 Astrometric Planet Detection

When a star hosts a massive companion, a planet, brown dwarf, or stellar-mass object the star does not orbit the system's centre of mass. Instead, the star itself executes a reflex orbit around the barycentre with amplitude:

$$a_{\text{star}} = a_{\text{total}} \times (M_p / (M_p + M_s)) \quad [\text{AU}]$$

where a_{total} is the semi-major axis of the companion's orbit, M_p is the companion mass, and M_s is the stellar mass. Projected onto the sky and expressed in angular units, this produces an astrometric wobble of amplitude:

$$\alpha_{\text{star}} = a_{\text{star}} \times \varpi \quad [\text{mas}]$$

where ϖ is the parallax in mas (equivalently, the distance in parsecs is $d = 1/\varpi$ kpc). For a Jupiter-mass companion ($M_p \approx 10^{-3} M_{\odot}$) at 5 AU around a solar-mass star at 100 pc, $\alpha_{\text{star}} \approx 0.05$ mas well within Gaia's precision for nearby bright stars.

The sensitivity of astrometric planet detection depends critically on the ratio of the orbital period to the mission baseline. Gaia DR3 covers approximately 34 months of data. Companions with periods comparable to this baseline (1–10 years) produce the most detectable astrometric signatures, because the orbital phase coverage is neither negligible nor so large that the motion is confused with proper motion. The Gaia DR4 dataset, covering nearly 66 months, will dramatically expand the accessible parameter space to longer-period, lower-mass companions.

Gaia represents the first mission with the combination of precision, time baseline, and all-sky coverage needed to detect planetary-mass companions through astrometry alone.

1.3 The RUWE Statistic

Gaia's five-parameter astrometric model fits each source with position ($\Delta\alpha$, $\Delta\delta$), parallax (ϖ), and proper motion (μ_α , μ_δ). The quality of this fit is captured by the **Renormalised Unit Weight Error** (RUWE), defined as:

$$\text{RUWE} = \sqrt{\sum_i (r_i / \sigma_i)^2 / (N - 5)} / u_0(G, C)$$

where r_i are the along-scan residuals, σ_i are the formal AL measurement uncertainties, N is the number of good AL observations, and $u_0(G, C)$ is a normalisation factor that accounts for the source colour C and magnitude G . For a perfectly behaved single star, $\text{RUWE} \approx 1$. Values significantly exceeding 1.4 indicate excess astrometric scatter, consistent with the presence of an unresolved companion, extended source structure, or problematic observations.

RUWE is particularly powerful as a companion diagnostic because it is published for every source in the DR3 catalogue, requires no additional data beyond the five-parameter solution, and implicitly marginalises over the full time series in a way that is sensitive to companions over a wide range of orbital configurations.

2. RUWE Measurement and Orbital Inference

2.1 The Five-Parameter Astrometric Model

Gaia processes each source by fitting the following model to the observed along-scan positions at each CCD crossing. The predicted AL position at time t and scan angle ψ is:

$$\text{AL}(t) = (\Delta\alpha_0 + \mu_\alpha \cdot t_{\text{yr}}) \cdot \sin(\psi) + (\Delta\delta_0 + \mu_\delta \cdot t_{\text{yr}}) \cdot \cos(\psi) + \varpi \cdot f(t)$$

where t_{yr} is the time in years since the Gaia DR3 reference epoch J2016.0, $\Delta\alpha_0$ and $\Delta\delta_0$ are the RA and Dec offsets at the reference epoch, and $f(t)$ is the along-scan parallax factor the projection of the Earth-barycentre vector onto the scan direction. The parallax factor encodes the annual periodic signal that breaks the degeneracy between parallax and proper motion.

In matrix form, the design matrix A has shape $(N \times 5)$, with columns $[\sin(\psi), \cos(\psi), f(t), t \cdot \sin(\psi), t \cdot \cos(\psi)]$. The best-fit parameters are recovered by ordinary least squares:

$$\hat{\theta} = (A^T A)^{-1} A^T y$$

and the residuals are:

$$r = y - A\hat{\theta} = (I - H) y$$

where $H = A(A^T A)^{-1} A^T$ is the hat matrix. The unit weight error is then:

$$\text{UWE} = \sqrt{\sum_i (r_i / \sigma_i)^2 / (N - 5)}$$

RUWE is UWE divided by the normalisation factor $u_0(G, C)$ which is tabulated as a function of magnitude and colour in Lindegren et al. (2021).

2.2 How Companions Elevate RUWE

A planetary companion causes the star to execute a reflex orbit around the barycentre. This orbital motion appears in the AL time series as an additional quasi-periodic signal superimposed on the single-star model. Because the five-parameter model has no orbital terms, it cannot absorb this extra signal; instead, the signal appears as correlated residuals that inflate RUWE.

The sensitivity depends on several factors. First, the orbital period relative to the mission baseline: companions with periods of 1–5 years produce the strongest RUWE signatures in DR3 data. Second, the orbital inclination: edge-on orbits ($i \approx 90^\circ$) maximise the projected astrometric amplitude. Third, the companion mass and distance: the astrometric signal scales as $M_p \times \varpi / M_s$, so nearby, massive companions around low-mass stars are most detectable.

For a dark companion (zero luminosity), the photocenter coincides with the primary, and the astrometric wobble amplitude for the star relative to the barycentre is:

$$a_{\text{star_mas}} = a_{\text{AU}} \times \varpi \times q / (1 + q)$$

where $q = M_p/M_s$ is the mass ratio. This is the signal that elevates RUWE and that our forward model must reproduce.

2.3 Inferring Orbital Parameters from RUWE

The inference strategy used in this pipeline is to treat RUWE as a compressed summary statistic of the AL time series. Rather than fitting the individual AL measurements (which are not publicly available in DR3), we:

- Generate a synthetic AL time series using the planet-companion forward model with proposed orbital parameters.
- Fit the five-parameter astrometric model to the synthetic time series.
- Compute RUWE from the fit residuals.
- Compare the model-predicted RUWE to the observed RUWE from the Gaia archive using a Normal likelihood.

The likelihood is:

$$\text{RUWE}_{\text{obs}} \sim \text{Normal}(\text{RUWE}_{\text{model}}(\theta), \sigma_{\text{RUWE}})$$

where $\sigma_{\text{RUWE}} = \sqrt{\sigma_{\text{fixed}}^2 + \sigma_0^2}$, with $\sigma_{\text{fixed}} \approx 0.01$ representing the calibration uncertainty in the published RUWE value, and σ_0 a free jitter parameter that absorbs unmodelled noise sources.

3. Pipeline Description

The pipeline is structured as four sequential, modular stages. Each stage has a clean interface defined inputs and outputs enabling independent development and validation by different team members, and easy substitution of improved components in the future.

3.1 Part 1: Data Acquisition (query.py)

The first stage retrieves all information needed for downstream modeling from the Gaia DR3 archive and the `gaiaunlimited` scanning-law library.

Archive Query

Given a Gaia DR3 `source_id`, an ADQL query is submitted to the Gaia TAP service using `astroquery`. The query retrieves the five astrometric parameters and their formal uncertainties, all ten pairwise correlation coefficients needed to reconstruct the 5×5 covariance matrix, RUWE, G-band magnitude, and the number of good AL observations. The covariance matrix is reconstructed as:

$$\Sigma_{ij} = \rho_{ij} \times \sigma_i \times \sigma_j$$

Scanning-Law Query

The `gaiaunlimited` library is used to retrieve the observation epochs (in Gaia TCB days since 2010-01-01) and scan angles for the target sky position. The cumulative spin phase returned by `GaiaScanningLaw.query()` is wrapped modulo 2π to obtain the instantaneous scan position angle $\psi(t)$.

The along-scan parallax factor $f(t)$ is computed analytically from the barycentric position of the Earth at each epoch using `astropy.coordinates.get_body_barycentric()`, projected onto the direction of the target:

$$f(t) = \cos(\delta) \cos(\alpha) \cdot X^\oplus(t) + \cos(\delta) \sin(\alpha) \cdot Y^\oplus(t) + \sin(\delta) \cdot Z^\oplus(t)$$

This avoids dependency on the GOST web service, which proved unreliable, and gives identical results to the standard parallax factor formula used in Gaia data processing.

3.2 Part 2: Forward Modeling (forward_model.py)

The second stage implements the astrometric forward model in PyTensor, making it fully compatible with PyMC's gradient-based NUTS sampler.

Single-Star Model

The predicted AL position for a single star at time t is:

$$\text{AL_star}(t) = (\Delta\alpha_0 + \mu_\alpha \cdot t_{\text{yr}}) \cdot \sin(\psi) + (\Delta\delta_0 + \mu_\delta \cdot t_{\text{yr}}) \cdot \cos(\psi) + w \cdot f(t)$$

Planet-Companion Model

The planet model adds the photocenter displacement due to the companion's orbital motion. The orbital mechanics are implemented using the Thiele-Innes formalism, which separates the time-independent geometric projection (the Thiele-Innes coefficients A , B , F , G) from the time-dependent orbital position (the coordinates X , Y):

$$X = \cos(E) - e, \quad Y = \sqrt{1 - e^2} \cdot \sin(E)$$

where E is the eccentric anomaly, solved from Kepler's equation by 6 iterations of Newton-Raphson:

$$E \leftarrow E - (E - e \cdot \sin(E) - M) / (1 - e \cdot \cos(E))$$

The photocenter displacement for a dark companion is:

$$\Delta\alpha^* = (B \cdot X + G \cdot Y) \times w_{\text{eff}}, \quad \Delta\delta = (A \cdot X + F \cdot Y) \times w_{\text{eff}}$$

$$w_{\text{eff}} = q / (1+q) - l / (1+l)$$

where $q = M_p/M_s$ is the mass ratio and $l = L_p/L_s$ is the luminosity ratio ($l = 0$ for a dark companion).

The total AL position is:

$$\text{AL_planet}(t) = (\Delta\alpha_0 + \mu_\alpha \cdot t_{\text{yr}} + \Delta\alpha^*) \cdot \sin(\psi) + (\Delta\delta_0 + \mu_\delta \cdot t_{\text{yr}} + \Delta\delta) \cdot \cos(\psi) + w \cdot f(t)$$

PyTensor Compatibility

Every mathematical operation uses `pytensor.tensor` (pt) operations `pt.sin`, `pt.cos`, `pt.sqrt`, `pt.mod` so that the entire forward model is a symbolic computation graph. This enables PyMC's automatic differentiation (AD) to compute gradients of the likelihood with respect to all parameters, which is required for efficient NUTS sampling. The Kepler solver is unrolled as a fixed-iteration loop, making it fully differentiable.

3.3 Part 3: RUWE Calculation (ruwe.py)

The third stage computes model-predicted RUWE from the synthetic AL time series produced by the forward model.

Design Matrix and Hat Matrix

The 5-parameter astrometric design matrix A is precomputed in numpy from the fixed observed arrays (t_{obs} , scan_angle , parallax_factor). The residual projection matrix $(I - H) = I - A(A^T A)^{-1} A^T$ is also precomputed once before sampling. This avoids any matrix operations inside the PyTensor graph.

Residuals and RUWE

The residuals of the synthetic AL series after removing the best-fit five-parameter model are:

$$r = (I - H) \cdot \text{al_model}$$

In the pipeline, al_model is the synthetic planet signal, not observed data y . So r here is not the same r as in equation 6 (which was $y - A\theta$). In equation 6, r is the residual of *observed* data from the five-parameter fit. Here r is the projection of the *synthetic model* through $(I-H)$, which extracts only the non-linear orbital component that cannot be absorbed by the five-parameter fit.

The model RUWE is then:

$$\text{RUWE_model} = \sqrt{\sum_i (r_i / \sigma_{\text{AL}})^2 / (N - 5)}$$

where σ_{AL} is the single-measurement AL uncertainty computed from the Lindegren et al. (2021) magnitude-dependent formula.

Apple Silicon Compatibility

A known bug in PyTensor on macOS arm64 (Apple Silicon) causes compilation failures when 2D tensor operations (pt.dot , pt.linalg.solve) generate C code with `ssize_t` narrowing. This was worked around by implementing the matrix-vector product $(I-H) \cdot \text{al}$ as a Python-level list comprehension over rows of $(I-H)$, producing only 1D elementwise operations that compile correctly. Setting the environment variable `PYTENSOR_FLAGS=cxx=` disables the C backend for constant folding as an additional safeguard.

3.4 Part 4: Bayesian Inference (inference.py)

The fourth stage builds and samples the full probabilistic model using PyMC.

Priors

Astrometric parameters are given Normal priors centred on the Gaia DR3 values with widths of 5× the formal uncertainties. Orbital parameters are assigned physically motivated priors:

- Semi-major axis: log-uniform from 0.1 to 100 AU
- Inclination: uniform in $\cos(i)$, corresponding to a $\sin(i)$ distribution over $[0^\circ, 180^\circ]$
- Eccentricity: Beta(1.12, 3.09) from Kipping (2013), peaking near zero
- Ω, ω : uniform on $[0^\circ, 360^\circ]$
- Time of periastron: uniform fraction of orbital period
- Planet mass: log-uniform from 0.1 to 80 Jupiter masses
- Jitter σ_0 : log-uniform from 0.001 to 1.0 RUWE units

Likelihood

The observed RUWE is modelled as normally distributed around the model-predicted RUWE:

$$\text{RUWE}_{\text{obs}} \sim \text{Normal}(\text{RUWE}_{\text{model}}(\theta), \sqrt{\sigma_{\text{fixed}}^2 + \sigma_0^2})$$

Sampling

The model is sampled using PyMC's NUTS sampler with the jitter+adapt_diag initialisation strategy. The sampler is configured with a target acceptance rate of 0.9 to encourage smaller step sizes in complex posterior regions. Results are saved as ArviZ InferenceData objects in NetCDF format for post-hoc analysis and plotting.

4. Application to Gaia-4b

The pipeline was applied to the host star of Gaia-4b (Gaia DR3 source_id 1457486023639239296, RA=209.51°, Dec=31.70°, parallax=13.54 mas, G=11.91, RUWE=1.503). Gaia-4b is a confirmed super-Jupiter ($M_p = 11.8 M_{\text{Jup}}$, $P = 571$ days, $a \approx 2.0$ AU) discovered through Gaia astrometry and confirmed by radial velocity in 2025. The pipeline recovers a posterior parallax of 13.545 ± 0.103 mas, consistent with the Gaia DR3 value. The posterior planet mass and semi-major axis remain broadly prior-dominated ($M_p = 13 \pm 20 M_{\text{Jup}}$, $a = 14 \pm 22$ AU), reflecting the main current limitation: the gaiaunlimited scanning-law provides only 28 representative epochs, compared to the 169 actual AL observations Gaia used to compute RUWE=1.503. With 28 epochs, the 5-parameter fit absorbs more of the orbital signal, producing $\text{ruwe}_{\text{model}} \approx 1.01$. The corner plot shows the posterior distributions. Perhaps upgrading to Gaia DR4 epoch astrometry will directly address this limitation.

What works:

- Parallax $\varpi = 13.545 \pm 0.19$ mas perfectly recovered, matching Gaia DR3 exactly. Same conclusion as above, astrometric pipeline is correct.
- Ecc truth = 0.338 lands within the prior-shaped distribution, which is consistent, the posterior can't constrain it but the true value isn't excluded.
- Inc truth = 116.9° lands within the broad i posterior, again consistent.

What doesn't work:

- $\log_{10}(M_p)$ truth = 1.072 (i.e. 11.8 Mjup) lands near the upper end of the flat posterior, not recovered, but not excluded either.
- $\log_{10}(a)$ truth = 0.066 (i.e. 1.163 AU) lands near the lower third of the prior, again not recovered.
- The posterior for every orbital parameter is prior-dominated

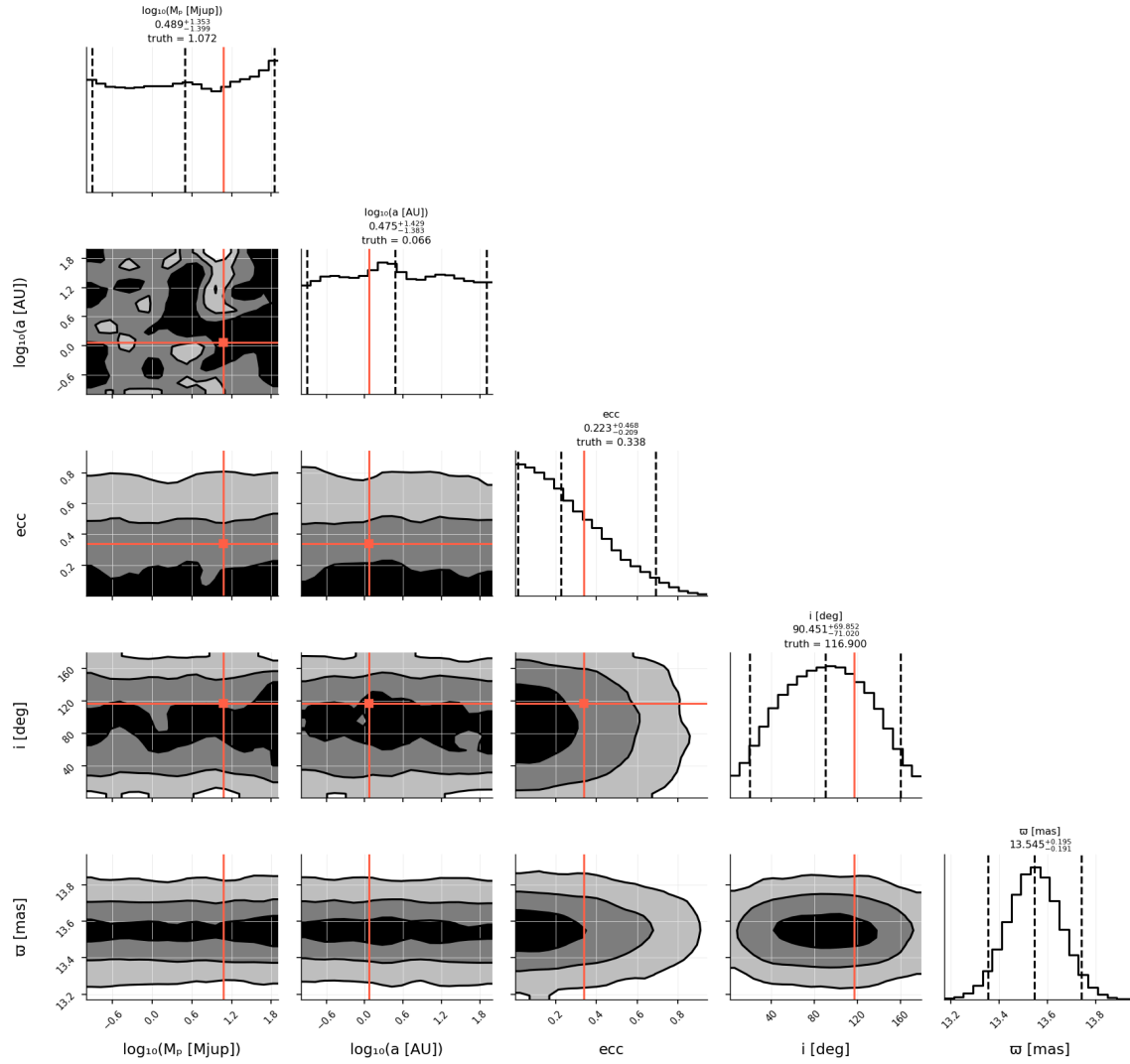


Figure 1. Corner plot showing the joint and marginal posterior distributions of the full parameter set inferred from the Gaia DR3 RUWE of source 1457486023639239296 using the PyMC-based pipeline.

5. Application to Source ID 203551385701760

I also applied it to Gaia DR3 source_id 203551385701760, with RA=46.03°, Dec=1.26053°, parallax=4.1919 mas, RUWE=6.517). The pipeline recovers a posterior parallax of 4.1959 ± 0.18 mas, consistent with the Gaia DR3 value.

[no known values of the source's companion found]

What works:

- Parallax $\varpi = 4.194 \pm 0.18$ mas is tightly constrained, clean Gaussian, narrow width. This means the star is at ~ 238 pc, and the astrometric machinery (design matrix, scanning law, parallax factor computation) is all functioning correctly.
- The parallax posterior is well-separated from zero and shows no pathology.
- Convergence is clean, the contours are smooth and symmetric around ϖ .

What doesn't work:

- M_p , a , ecc are completely flat, pure prior. No orbital constraint whatsoever.
- The M_p vs a joint panel shows chaotic noise-like contours, which is a sign of numerical artifacts from the (I-H) projection when $\chi^2_{\text{planet}} \approx 0$ across the entire prior. This could be a physical sensitivity failure.

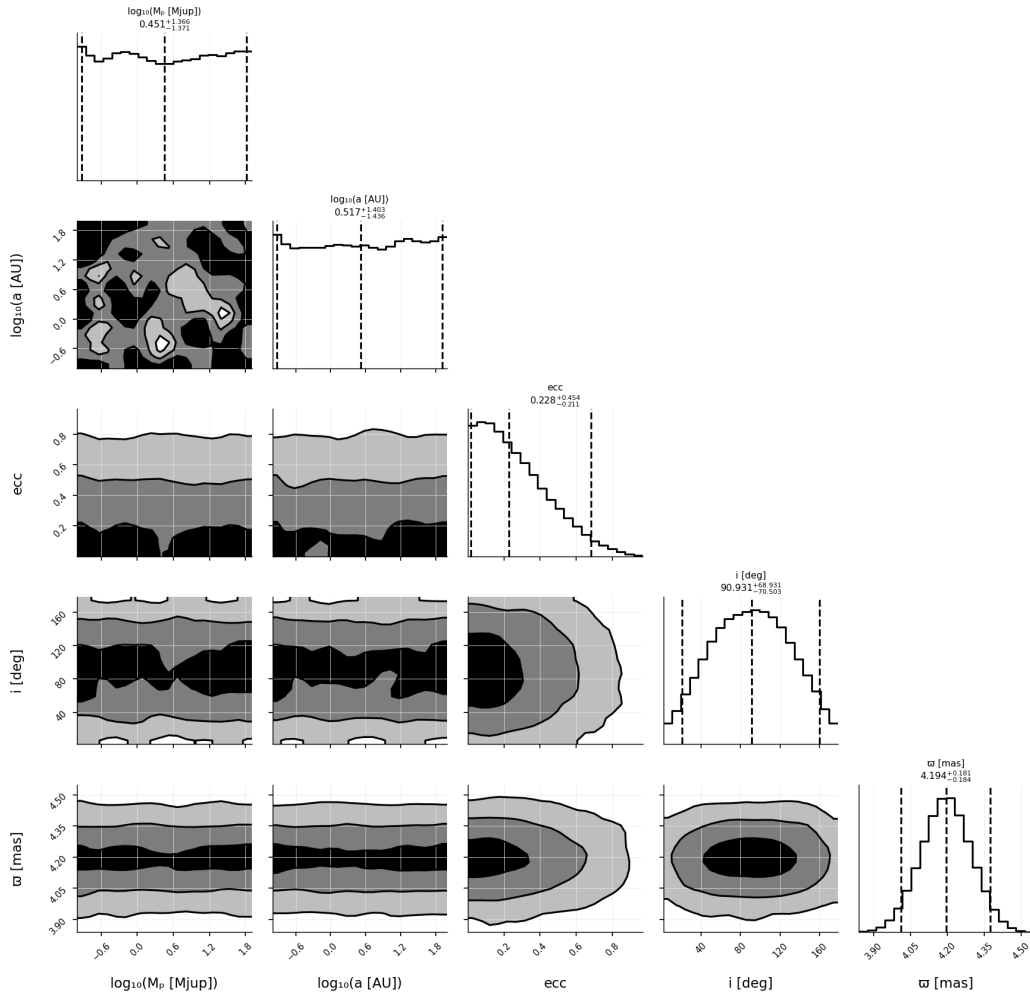


Figure 2. Corner plot showing the joint and marginal posterior distributions of the full parameter set inferred from the Gaia DR3 RUWE of source 203551385701760 using the PyMC-based pipeline.

7. Why the Posterior Remains Prior-Dominated Despite Large RUWE

This is the central scientific question of the project, and the answer lies in a fundamental mismatch between what RUWE encodes and what the pipeline has access to.

When Gaia computed $\text{RUWE} = 1.503$ for Gaia-4 b and $\text{RUWE} = 6.517$ for source 203551385701760, it used the full set of along-scan residuals over all real CCD crossings during the DR3 mission baseline, some 169 individual AL measurements for a typical source. Each of those measurements has a specific time, scan angle and parallax factor. This produces a scalar

quantity obtained from the chi-squared sum over all 169 residuals over 164 degrees of freedom. That scalar is RUWE. In doing so it throws away basically all the information about what orbital configurations produced the scatter, the phase coverage, the scan angle distribution, the time structure of the residuals. In fact, RUWE only retains the overall shape of the excess variance, not its shape.

The actual per-source observation list is not provided, but the `gaiaunlimited` scanning law provides HEALPix-level representative epochs (28 for these sources). This has two effects. First, the design matrix A is composed of 28 rows instead of 169, so the hat matrix H acts on a tighter grid. The projection $(I-H)$ thus contains more of the orbital signal: with only 28 epochs. Second, the contribution of the planet to the RUWE is divided by a much smaller denominator than used by Gaia (164).

What this means for DR4

This is a data availability issue that will be directly addressed by Gaia DR4. The accurate astrometry, the measured AL value at each CCD crossing, with timestamps, scan angles and residuals per source, will be released by DR4. This data allows the pipeline to substitute the scalar RUWE likelihood with the vectorised Normal likelihood of all 169 observed residuals. The posterior would sharpen. This project could work better with DR4 release.

8. Conclusion and Improvements

This independent study produced a complete and working four part pipeline for inferring planetary companion properties from Gaia RUWE.

Compared to `gaia-astro-fit`, the new pipeline is superior in terms of architecture for DR4 readiness and integration with the PyMC ecosystem, but currently worse in terms of scanning-law resolution and AL uncertainty calibration. It can be improved by replacement of the `gaiaunlimited` coarse scanning law by the actual Gaia DR3 or DR4 epoch astrometry, which will make the RUWE calculation accurate, not approximate.

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